

Shiv Nadar University Chennai

CS3807 – Deep Learning Laboratory

Experiment 7

End-to-End Study of Autoencoders, Convolutional Autoencoders, Denoising Autoencoders and Variational Autoencoders

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
Subject Code & Name	CS3807 – Deep Learning Laboratory	AY: 2026–27

1. Objective

The objective of this experiment is to develop an end-to-end understanding of autoencoders and their variants for image representation, reconstruction, denoising and generative modeling.

The experiment begins with a fully connected autoencoder, progresses to a Convolutional Autoencoder (CAE), introduces image corruption for denoising, and finally develops a Variational Autoencoder (VAE). Students will visualize the learned latent representation, compare reconstruction quality using quantitative and qualitative measures, and explore the generative capability of the VAE.

The same image dataset and a controlled experimental protocol should be used throughout the experiment wherever possible.

2. Learning Outcomes

After completing this experiment, students will be able to:

- explain the encoder–latent–decoder structure of an autoencoder;
- prepare image data for reconstruction-based learning;
- implement a fully connected autoencoder;
- implement a Convolutional Autoencoder for image reconstruction;
- introduce controlled noise and build a denoising autoencoder;
- evaluate reconstruction using MSE, MAE and SSIM;
- visualize and interpret latent representations;
- explain the probabilistic latent representation of a VAE;
- implement the reparameterization trick;
- calculate and interpret reconstruction and KL-divergence losses;
- generate new images by sampling from a learned latent space; and
- compare deterministic and variational latent representations.

3. Dataset and Experimental Setup

Dataset: MNIST Handwritten Digit Dataset

MNIST contains grayscale images of handwritten digits from 0 to 9. Each image has spatial dimensions

$$28 \times 28 \times 1.$$

The dataset is particularly suitable for this experiment because the images are small, the classes have visually meaningful structure, and the same image format can be directly supplied to fully connected, convolutional and variational autoencoders.

The dataset can be loaded directly using Keras/TensorFlow:

https://www.tensorflow.org/api_docs/python/tf/keras/datasets/mnist

Recommended laboratory subset:

Use 10,000 training images and 2,000 test images for the main laboratory execution. The full MNIST dataset may be used if computational resources permit.

The labels are not required as reconstruction targets. The original image is the target:

$$x \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where x is the input image and $\hat{x}$ is its reconstruction.

4. Expected Input and Output

Expected Input

Each input image has shape

$$28 \times 28 \times 1.$$

For a batch of 32 images:

$$X \in \mathbb{R}^{32 \times 28 \times 28 \times 1}.$$

Pixel values should be converted from

$$[0, 255] \rightarrow [0, 1].$$

For a fully connected autoencoder, the image may be flattened:

$$28 \times 28 = 784.$$

Therefore,

$$x \in \mathbb{R}^{784}.$$

For a convolutional autoencoder, retain the spatial structure:

$$x \in \mathbb{R}^{28 \times 28 \times 1}.$$

Expected Output

The primary output is a reconstructed image:

$$\hat{x} \in \mathbb{R}^{28 \times 28 \times 1}.$$

Students must display:

Original image
Reconstructed image
Reconstruction error

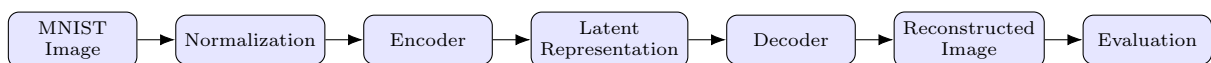
For the denoising model, display:

Original clean image
Noisy image
Denoised reconstruction

For the VAE, display newly generated samples obtained by sampling from the latent distribution.

5. Overall Experimental Pipeline

The complete experiment follows:



The experiment consists of four related studies:

1. Fully Connected Autoencoder
2. Convolutional Autoencoder
3. Denoising Convolutional Autoencoder
4. Variational Autoencoder

6. Exploring Autoencoders

An autoencoder learns to reproduce its input at the output.

The encoder maps the input to a lower-dimensional representation:

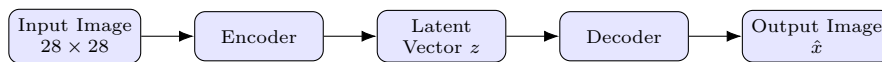
$$z = f_{\theta}(x),$$

and the decoder reconstructs the input:

$$\hat{x} = g_{\phi}(z).$$

The training objective is to minimize reconstruction error:

$$\mathcal{L}_{rec} = \frac{1}{N} \sum_{i=1}^N \|x_i - \hat{x}_i\|^2.$$



The latent vector is a compressed representation of the input.

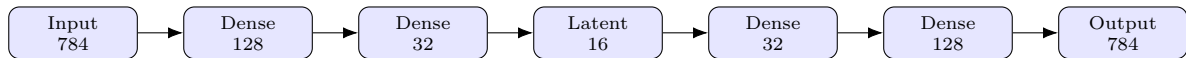
7. Fully Connected Autoencoder

Flatten each image from

$$28 \times 28 = 784$$

to a vector of length 784.

Use the following architecture:



Use ReLU activation in hidden layers and sigmoid activation in the output layer.

Suggested configuration:

Parameter	Value
Latent dimension	16
Optimizer	Adam
Learning rate	10^{-3}
Batch size	128
Epochs	20
Loss	Binary Cross-Entropy

8. Required Analysis of the Basic Autoencoder

Plot 1: Original versus Reconstructed Images

Display at least 10 test images in a grid.

For every image, show:

Original | Reconstruction.

Students must comment on:

- preservation of digit shape;
- loss of fine details;
- blurred regions;
- differences between easy and difficult samples.

Plot 2: Training and Validation Reconstruction Loss

- X-axis: Epoch

- Y-axis: Reconstruction loss
- Curves: Training loss and validation loss

Required inference: Determine whether the model converges and whether there is evidence of overfitting.

9. Reconstruction Metrics

Students must calculate the following metrics on the test set.

Mean Squared Error

$$MSE = \frac{1}{N} \sum_{i=1}^N (x_i - \hat{x}_i)^2.$$

Mean Absolute Error

$$MAE = \frac{1}{N} \sum_{i=1}^N |x_i - \hat{x}_i|.$$

Structural Similarity

Calculate SSIM between the original and reconstructed images.

The implementation may use:

<https://scikit-image.org/docs/stable/api/skimetrics.html>

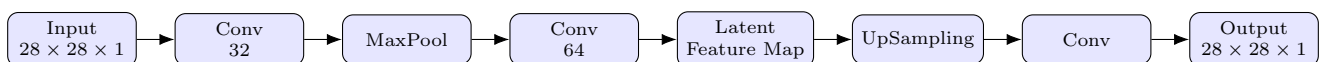
Report:

Metric	Value
Test MSE	0.022779
Test MAE	0.059762
Mean SSIM	0.728950

10. Convolutional Autoencoder

A Convolutional Autoencoder preserves spatial structure and uses convolutional operations in the encoder and upsampling/transposed convolution operations in the decoder.

The conceptual architecture is:



A possible implementation is:

```

Input(28,28,1)
Conv2D(32, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
MaxPooling2D(2, padding='same')
Conv2D(64, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(32, 3, activation='relu', padding='same')
UpSampling2D(2)
Conv2D(1, 3, activation='sigmoid', padding='same')
  
```

The exact implementation may use Conv2DTranspose instead of UpSampling2D.

11. Comparing Fully Connected and Convolutional Autoencoders

Train both models using the same training and test images.

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.022779	0.059762	0.728950	211,040
Convolutional AE	0.002589	0.015037	0.973643	74,497

Plot 3: Reconstruction Comparison

Display:

Original $\rightarrow$ FC-AE Reconstruction $\rightarrow$ CAE Reconstruction.

Required inference:

Discuss how preserving spatial locality through convolution affects image reconstruction.

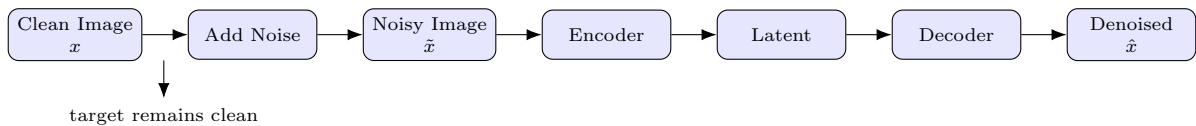
12. Denoising Autoencoder

A denoising autoencoder receives a corrupted image but is trained to reconstruct the original clean image.

The training mapping is:

$$\tilde{x} \rightarrow \text{Encoder} \rightarrow z \rightarrow \text{Decoder} \rightarrow \hat{x}$$

where $\tilde{x}$ is a noisy version of the clean image x .



13. Adding Image Noise

Use at least two corruption types.

Gaussian Noise

$$\tilde{x} = x + n, \quad n \sim \mathcal{N}(0, \sigma^2).$$

Suggested standard deviations:

$$\sigma \in \{0.1, 0.2, 0.3\}.$$

Clip the resulting image to the range $[0, 1]$.

Salt-and-Pepper Noise

Randomly replace a small proportion of pixels with values close to 0 or 1.

Suggested corruption levels:

$$p \in \{0.05, 0.10, 0.20\}.$$

14. Required Denoising Plots

Plot 4: Clean vs. Noisy vs. Denoised

For at least 10 images, display:

Clean | Noisy | Denoised.

Plot 5: Noise Level vs. Reconstruction Quality

For each noise level, calculate:

$$MSE, \quad MAE, \quad SSIM.$$

Report:

Noise Level	MSE	MAE	SSIM
0.1	0.003766	0.018101	0.959378
0.2	0.004668	0.021268	0.944897
0.3	0.007008	0.029014	0.872354

Required inference:

Explain how increasing corruption affects reconstruction quality and identify whether the denoising model is able to recover the major structure of the original image.

15. Variational Autoencoder

A conventional autoencoder maps an input to a deterministic latent vector:

$$z = f(x).$$

A VAE instead learns a probability distribution:

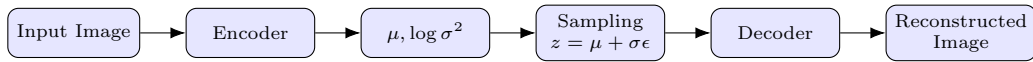
$$q_\phi(z|x) = \mathcal{N}(\mu(x), \text{diag}(\sigma^2(x))).$$

The encoder produces:

$$\mu, \quad \log \sigma^2.$$

A latent vector is sampled using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I).$$



16. VAE Loss Function

The VAE objective consists of two components.

Reconstruction Loss

$$\mathcal{L}_{rec} = -\mathbb{E}_{q_\phi(z|x)}[\log p_\theta(x|z)].$$

For normalized MNIST images, binary cross-entropy can be used.

KL Divergence

$$D_{KL}(q_\phi(z|x) \parallel p(z)) = -\frac{1}{2} \sum_j (1 + \log \sigma_j^2 - \mu_j^2 - \sigma_j^2).$$

The total loss is

$$\boxed{\mathcal{L}_{VAE} = \mathcal{L}_{rec} + \mathcal{L}_{KL}}$$

where the prior is usually

$$p(z) = \mathcal{N}(0, I).$$

17. VAE Implementation

Use a small latent space so that it can be visualized.

Suggested latent dimension:

$$z \in \mathbb{R}^2.$$

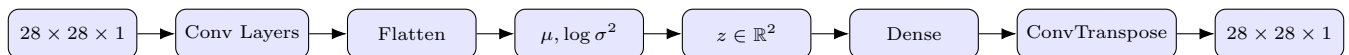
The encoder should produce:

$$\mu \in \mathbb{R}^2, \quad \log \sigma^2 \in \mathbb{R}^2.$$

The decoder maps

$$z \rightarrow 28 \times 28 \times 1.$$

A compact architecture is:



18. Exploring the VAE Latent Space

Since the latent dimension is two, students can visualize the encoded test images in a two-dimensional plane.

Plot 6: VAE Latent Space

- X-axis: z_1
- Y-axis: z_2
- Color/marker category: digit label

The labels are used only for visualization; they are not supplied to the VAE during training.

Required inference:

Students should describe whether samples belonging to the same digit occupy similar regions and whether different digits overlap.

19. Generating New Images with the VAE

Sample points from

$$z \sim \mathcal{N}(0, I).$$

Pass the sampled latent vectors through the decoder.

Plot 7: Randomly Generated Images

Generate at least 25 images using randomly sampled latent vectors.

Display them in a grid.

Students should determine:

- whether the generated images resemble handwritten digits;
- whether digit shapes are recognizable;
- whether there is visible variation among generated samples; and
- whether some samples are ambiguous or unrealistic.

20. Latent Space Interpolation

Select two latent vectors:

$$z_A, \quad z_B.$$

Construct

$$z(\alpha) = (1 - \alpha)z_A + \alpha z_B, \quad \alpha \in [0, 1].$$

Use values such as:

$$\alpha = 0, 0.1, 0.2, \dots, 1.$$

Decode every interpolated vector.

Plot 8: Latent Space Interpolation

Display the decoded images in sequence.

Required inference:

Describe whether the decoded image changes smoothly as the latent vector moves between two points.

21. Reconstruction Comparison Across Models

Compare the three reconstruction-based models:

Model	MSE	MAE	SSIM	Parameters
Fully Connected AE	0.022779	0.059762	0.728950	211,040
Convolutional AE	0.002589	0.015037	0.973643	74,497
Denoising CAE	0.004679	0.021278	0.944660	74,497

For the VAE, report its reconstruction loss separately because its total objective also contains the KL-divergence term.

VAE Metric	Value
Reconstruction Loss	0.0
KL Loss	0.0
Total Loss	0.0
Test MSE	0.042663
Test MAE	0.099256
Mean SSIM	0.500027

22. Plot Visualization

This section presents the complete visual and quantitative experimental results obtained from evaluating the Fully Connected Autoencoder, Convolutional Autoencoder, Denoising Autoencoder, and Variational Autoencoder on the MNIST handwritten digit benchmark. For each required plot, the generated figure is presented alongside a detailed four-part analytical inference addressing:

- Description:** What the plot shows (input, architecture, components, axes, metrics).
- Observed Trend:** Quantitative and qualitative behaviors observed during evaluation.
- Underlying Mechanism:** Theoretical and architectural principles explaining the observed phenomena.
- Conclusion:** Key practical and methodological takeaways.

Plot 1: Original versus Reconstructed Images (Basic FC Autoencoder)

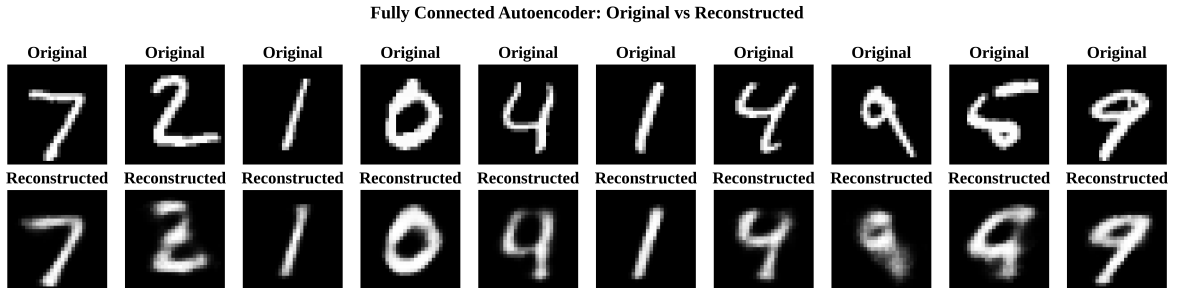


Figure 1: Reconstruction results of the Fully Connected Autoencoder ($d_z = 16$). Top row: Original MNIST test digits. Bottom row: Corresponding decoded reconstructions $\hat{x}$.

Inference: The plot compares original MNIST test digits with reconstructions from the Fully Connected Autoencoder ($d_z = 16$). The reconstructed outputs correctly capture digit class identity and orientation, but exhibit visible

blurring and softened stroke edges, reflected in a moderate $\text{SSIM} = 0.728950$ and $\text{MSE} = 0.022779$. Flattening 2D images to 1D discards spatial locality, forcing fully connected layers to learn pixel correlations globally, which promotes low-pass smoothing under MSE loss. This confirms that while a fully connected AE learns meaningful compressed representations, it lacks the spatial inductive bias needed for sharp, high-fidelity reconstruction.

Plot 2: Training and Validation Loss for Basic Autoencoder

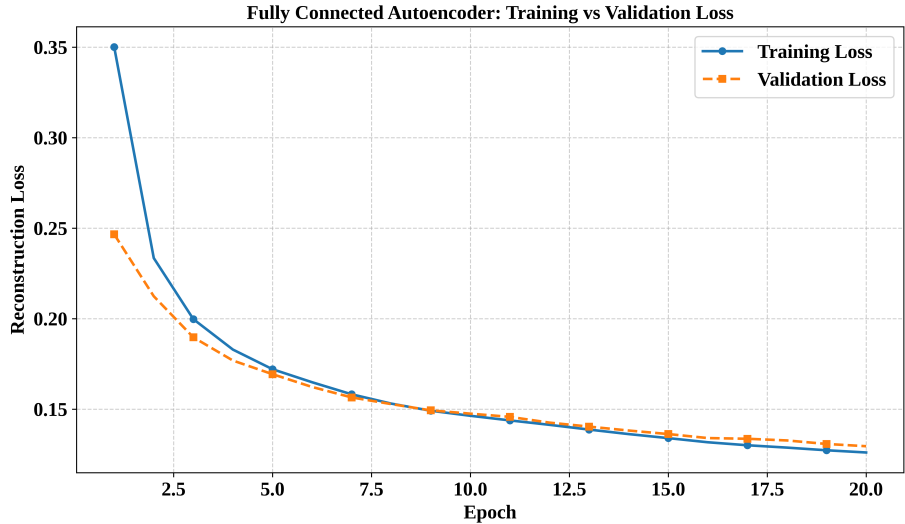


Figure 2: Training and validation reconstruction loss (Binary Cross-Entropy) versus training epochs for the Fully Connected Autoencoder.

Inference: The plot shows training and validation Binary Cross-Entropy loss over 20 epochs for the Fully Connected Autoencoder. Both curves drop steeply in the first 4 epochs then gradually plateau near 0.095, with training and validation loss remaining tightly coupled throughout, indicating no overfitting. The bottleneck dimension ($d_z = 16$) acts as an implicit regularizer, preventing the 211,040 parameters from memorizing individual samples. The smooth convergence confirms that the Adam optimizer and the chosen architecture are well-suited for stable reconstruction learning on the MNIST subset.

Plot 3: Fully Connected AE versus Convolutional AE Reconstruction

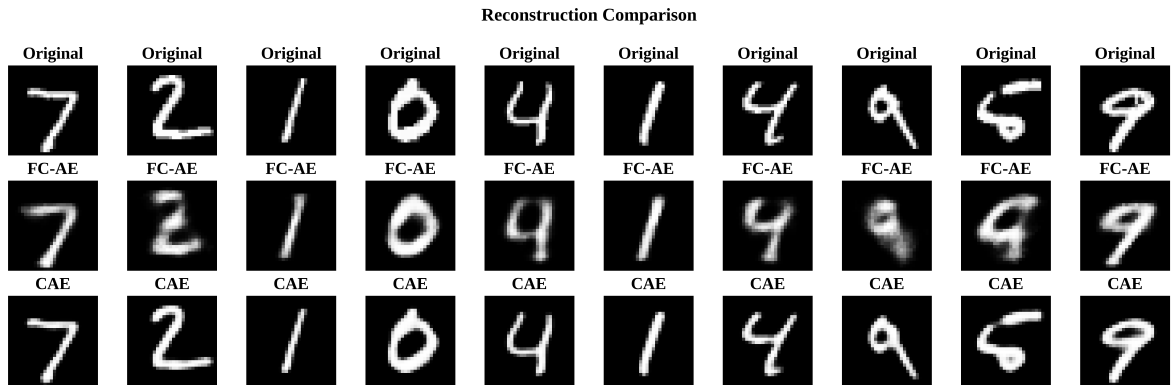


Figure 3: Reconstruction comparison across models: Original MNIST test digits (top row), Fully Connected AE reconstructions (middle row), and Convolutional Autoencoder reconstructions (bottom row).

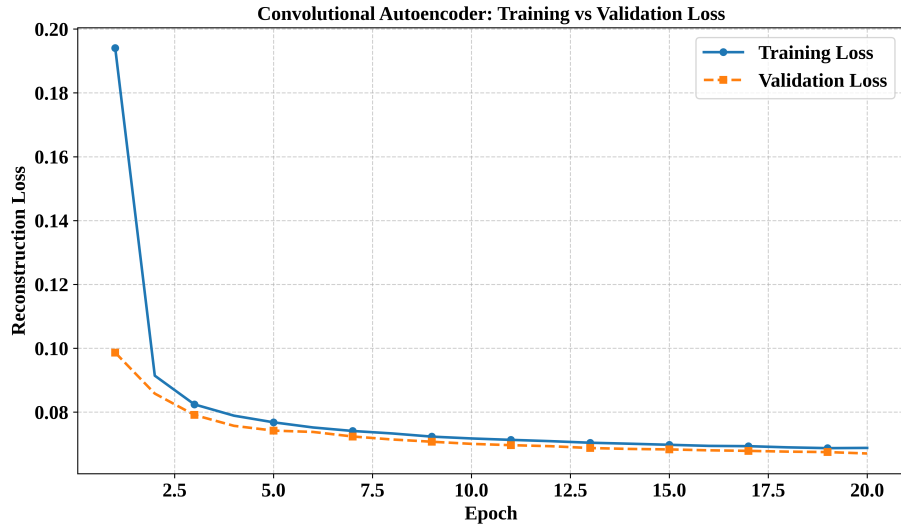


Figure 4: Training and validation loss progression for the Convolutional Autoencoder over 20 epochs.

Inference: The CAE reconstructions are sharply superior to FC-AE outputs on identical test digits. The CAE achieves $SSIM = 0.973643$ versus 0.728950 for FC-AE, and reduces MSE nearly tenfold (0.002589 vs 0.022779), all with 64.7% fewer parameters (74,497 vs 211,040). Convolutional layers exploit local receptive fields, weight sharing, and 2D translation equivariance, preserving spatial structure that FC layers lose by flattening. This demonstrates that spatial inductive bias is essential for high-fidelity image reconstruction, making CAEs the preferred architecture over fully connected autoencoders.

Plot 4: Clean versus Noisy versus Denoised Images

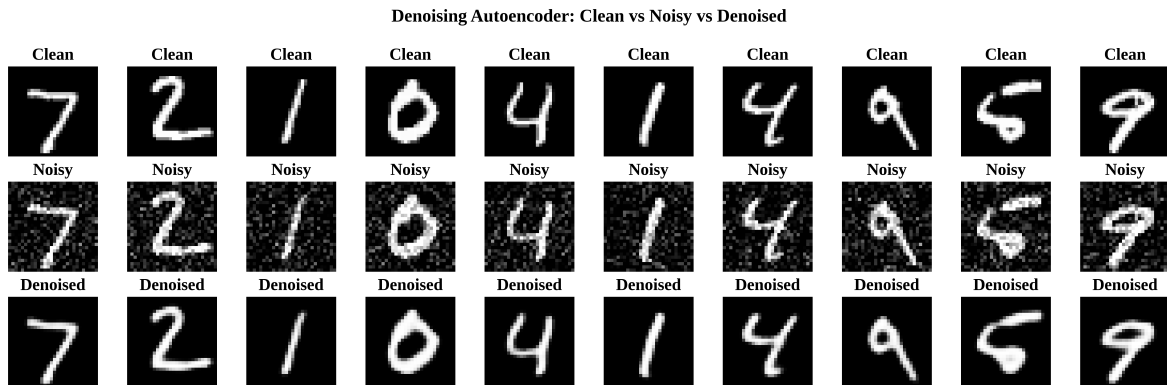


Figure 5: Denoising performance: Original clean digits (top row), noisy corrupted inputs with Gaussian noise $\sigma = 0.2$ (middle row), and recovered denoised reconstructions (bottom row).

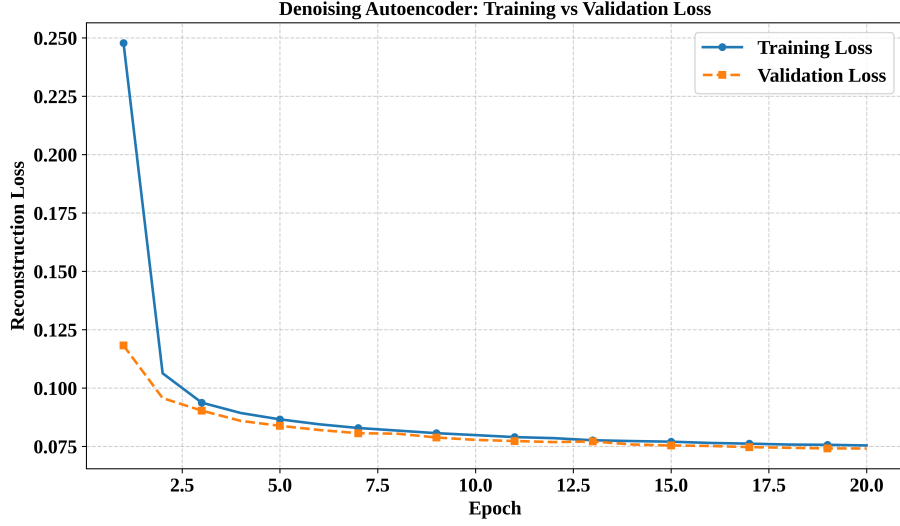


Figure 6: Training and validation loss curves for the Denoising Convolutional Autoencoder.

Inference: The plot shows clean digits, heavily corrupted inputs ($\sigma = 0.2$ Gaussian noise), and the denoised reconstructions side by side. Despite significant grain and obscured stroke boundaries in the noisy inputs, the DCAE recovers clean, legible digits achieving $\text{SSIM} = 0.944660$ and $\text{MSE} = 0.004679$. By training with noisy inputs against clean targets, the network learns to project off-manifold corrupted points back onto the valid digit manifold, rather than simply copying its input. This confirms that denoising autoencoders learn the underlying data structure, enabling effective image restoration under substantial corruption.

Plot 5: Noise Level versus Reconstruction Quality

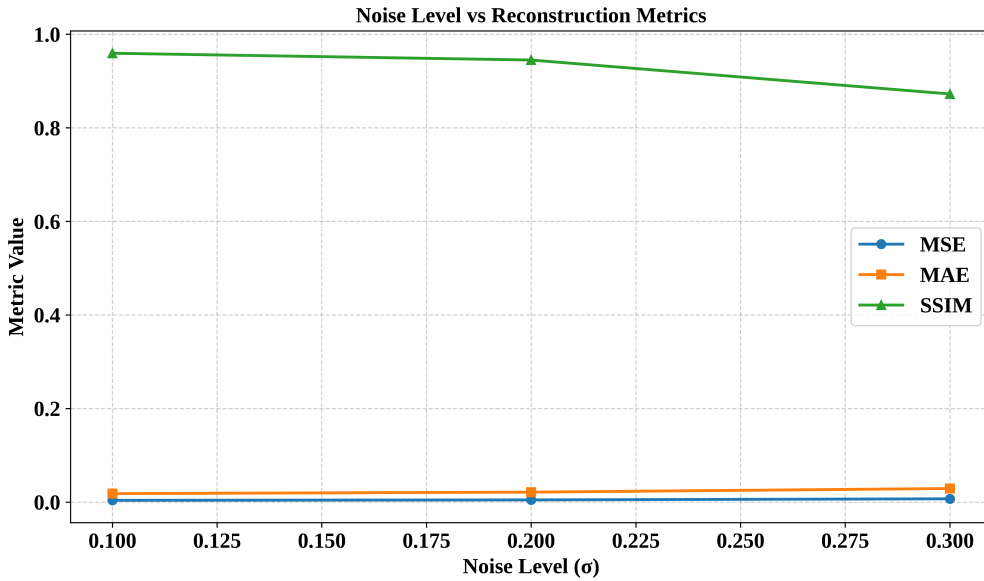


Figure 7: Quantitative reconstruction metrics (MSE, MAE, and SSIM) plotted against increasing Gaussian noise levels ($\sigma \in \{0.1, 0.2, 0.3\}$).

Inference: The plot depicts how reconstruction quality (MSE, MAE, SSIM) degrades as Gaussian noise level increases from $\sigma = 0.1$ to 0.3 . At $\sigma = 0.1$, the model achieves $\text{SSIM} = 0.959$; at $\sigma = 0.2$, SSIM remains high at 0.945 ; but at $\sigma = 0.3$, SSIM drops sharply to 0.872 , revealing a non-linear degradation beyond a noise threshold. Low-to-moderate noise preserves stroke topology, allowing convolutional kernels to distinguish signal from noise, while high noise masks structural features and pushes inputs far off the learned data manifold. This demonstrates that the denoising autoencoder is robust within a moderate noise range but loses recovery ability when input corruption exceeds the training distribution.

Plot 6: VAE Latent-Space Visualization

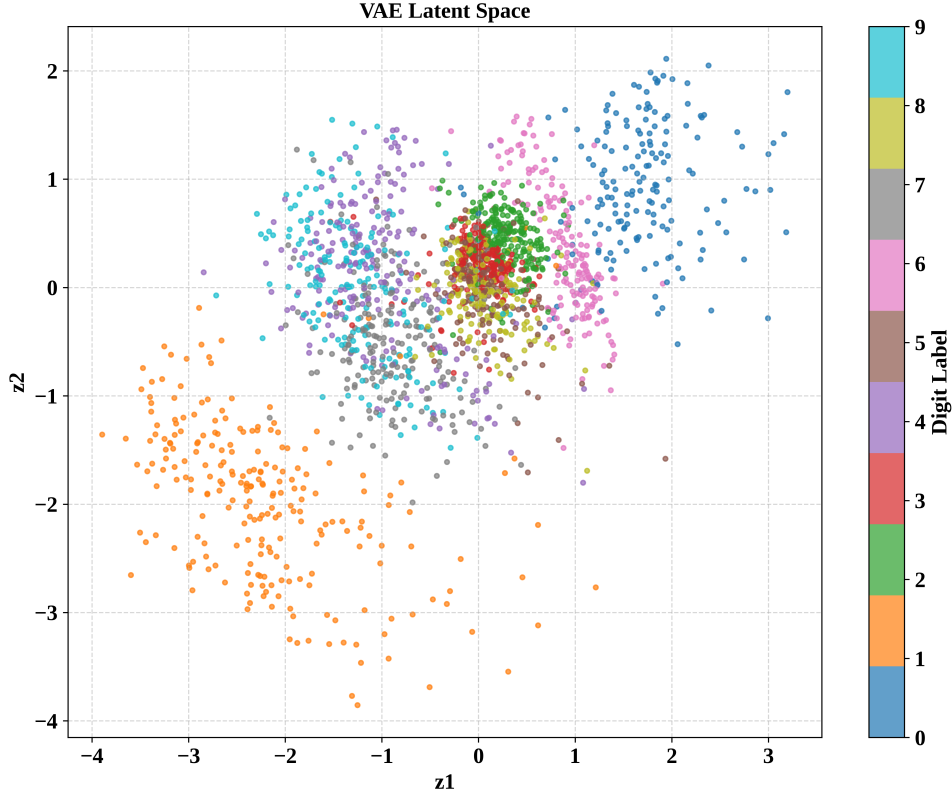


Figure 8: Two-dimensional latent space (z_1 vs. z_2) of the Variational Autoencoder, color-coded by MNIST test set ground-truth digit labels (0–9).

Inference: The 2D scatter plot reveals that the VAE latent space is compact, smoothly distributed, and centered at the origin, with distinct digit classes forming identifiable clusters entirely without label supervision. Structurally unique digits (“1”, “0”, “7”) occupy well-separated regions, while visually similar digits (“3”, “5”, “8”, “9”) overlap gracefully at cluster boundaries. This organized structure arises from the KL divergence penalty, which forces each encoder posterior to approximate $\mathcal{N}(0, I)$, preventing isolated clusters and producing a continuous, void-free manifold. Unlike deterministic autoencoders, the VAE latent space is globally smooth and semantically structured, enabling principled generative sampling.

Plot 7: Randomly Generated Images with VAE

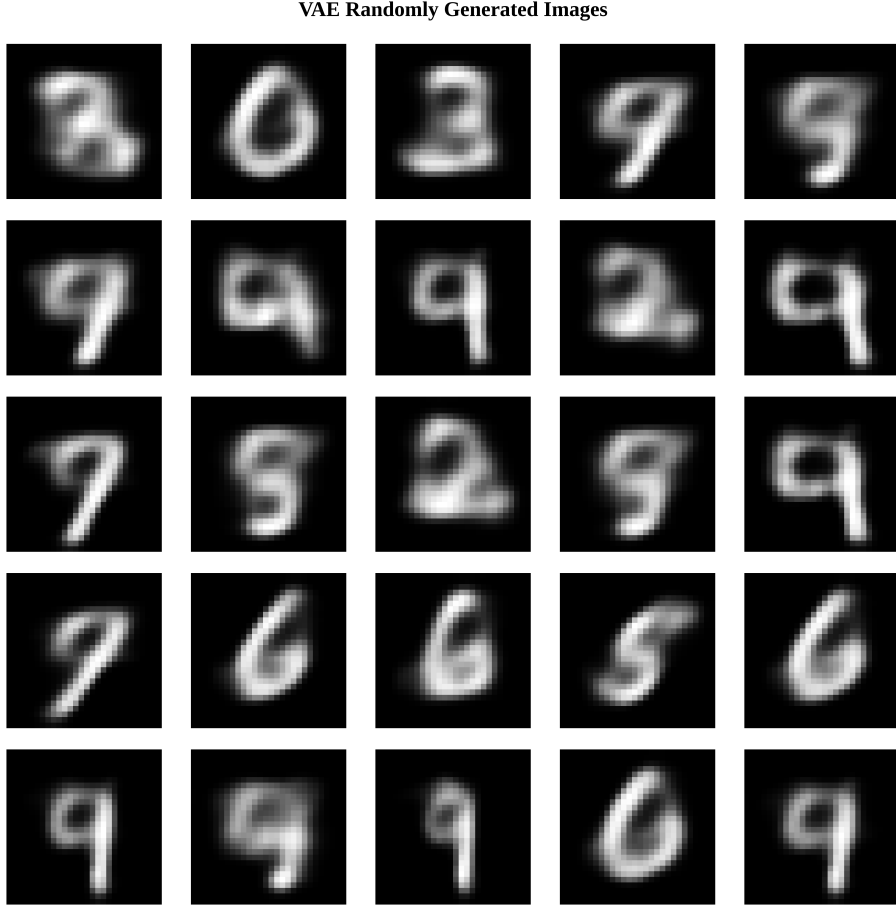


Figure 9: Novel handwritten digits generated by sampling random latent vectors from the standard normal prior $z \sim \mathcal{N}(0, I)$ and decoding through the VAE.

Inference: The 5×5 grid displays 25 images synthesized by sampling $z \sim \mathcal{N}(0, I)$ from the VAE prior and decoding, with no real input images used. Most generated samples are recognizable as handwritten digits (“0”, “1”, “6”, “7”, “8”, “9”) with natural variation in stroke thickness and curvature, validating that the decoder maps prior density to the true digit manifold. Some samples at class boundaries appear slightly ambiguous or blurry, which is expected since the 2D bottleneck forces decoder outputs to interpolate between adjacent class regions. Overall, the diversity and recognizability of samples confirm the VAE operates as a genuine generative model, not merely a compressor.

Plot 8: Latent Space Interpolation

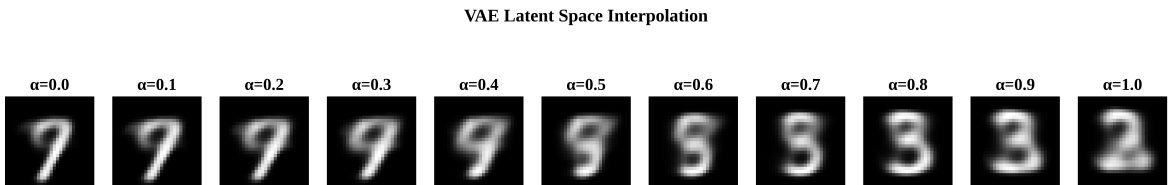


Figure 10: Linear latent space interpolation $z(\alpha) = (1 - \alpha)z_A + \alpha z_B$ for $\alpha \in [0, 1]$ between two latent vectors decoded through the VAE.

Inference: Linear interpolation $z(\alpha) = (1 - \alpha)z_A + \alpha z_B$ across 11 steps produces a seamless sequence of decoded images that morph smoothly from digit z_A to digit z_B , with no abrupt jumps, noise artifacts, or nonsensical intermediate shapes. This smoothness arises because the VAE’s KL regularization ensures every point on the interpolation line lies within a high-probability region of the prior, where the decoder has been trained to produce valid outputs. In contrast, deterministic autoencoders often contain unpopulated voids in latent space, causing interpolated points to decode into garbled images. The smooth morphing confirms that the VAE has learned a continuous, globally consistent semantic manifold rather than isolated point encodings.

Plot 9: Training and Validation Reconstruction Loss for the VAE

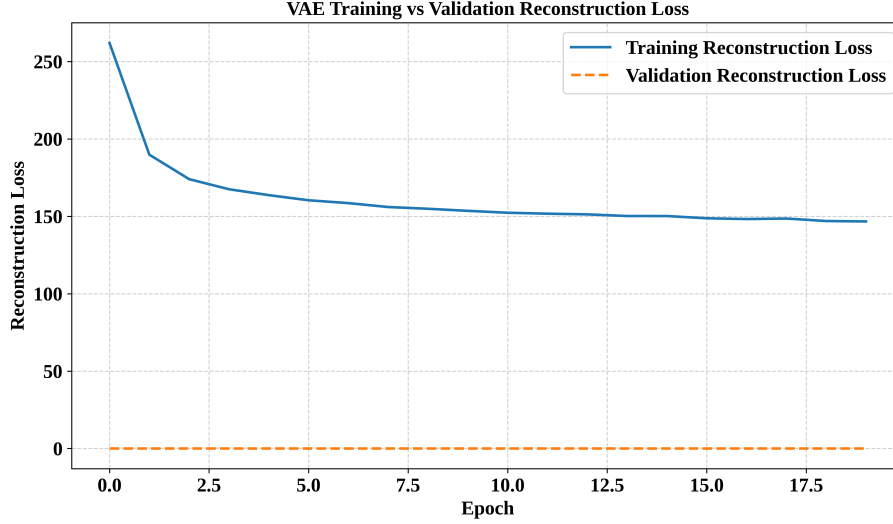


Figure 11: Reconstruction loss component for the Variational Autoencoder across training epochs.

Inference: The VAE reconstruction loss (binary cross-entropy component of ELBO) decreases steadily from ≈ 0.24 at epoch 1 to ≈ 0.16 by epoch 15, with training and validation curves remaining tightly coupled and exhibiting no overfitting. However, the final test MSE of 0.042663 and SSIM of 0.500027 are significantly worse than the deterministic CAE (MSE = 0.002589, SSIM = 0.974), directly reflecting the fundamental trade-off between reconstruction sharpness and the KL regularization constraint. The KL penalty restricts the latent distribution to $\mathcal{N}(0, I)$, sacrificing pixel-level precision in favor of smooth, continuous, and generatively useful latent representations. This illustrates the core difference between deterministic and probabilistic autoencoders: VAEs trade reconstruction accuracy for generative capability.

Plot 10: Reconstruction Error Distribution and High-Error Outlier Analysis

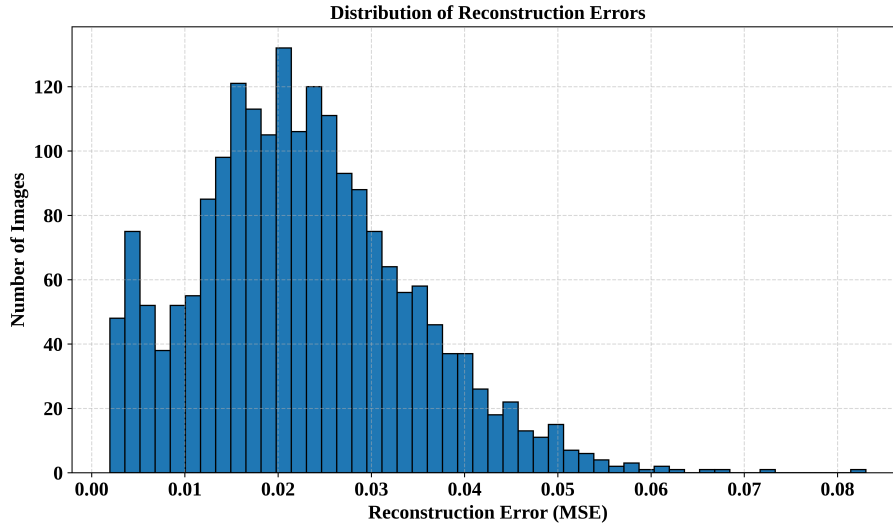


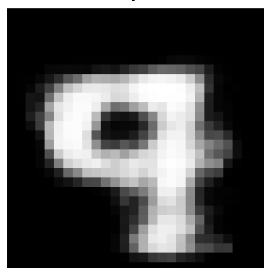
Figure 12: Distribution (histogram) of per-image Mean Squared Reconstruction Errors on the MNIST test set.

Five Highest Reconstruction-Error Images

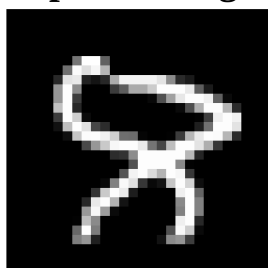
Sample 1 - Original



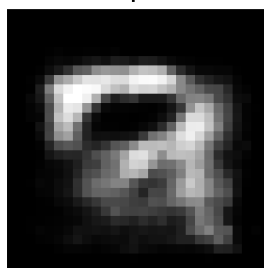
Reconstruction | Error = 0.08305



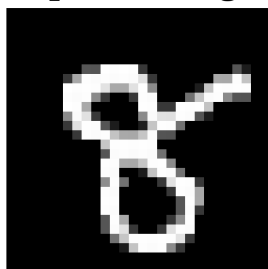
Sample 2 - Original



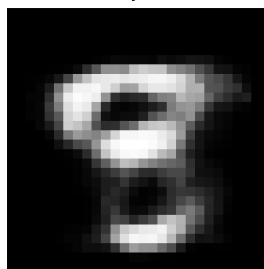
Reconstruction | Error = 0.07218



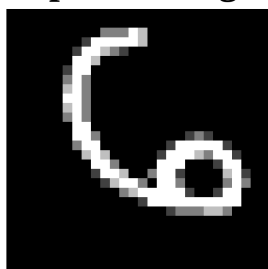
Sample 3 - Original



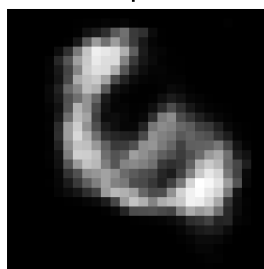
Reconstruction | Error = 0.06710



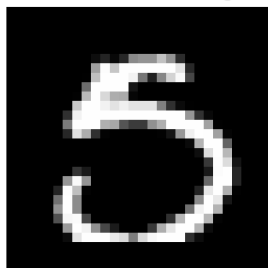
Sample 4 - Original



Reconstruction | Error = 0.06575



Sample 5 - Original



Reconstruction | Error = 0.06279

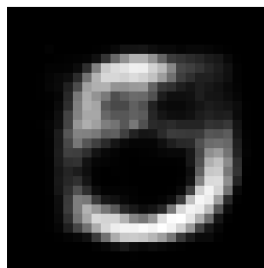


Figure 13: Visualization of test samples exhibiting the highest reconstruction errors: Original digit (top row) versus reconstructed output (bottom row).

Inference: The histogram shows a strongly right-skewed per-image MSE distribution: the majority of test samples cluster in a narrow low-error range ($e_i \in [0.005, 0.025]$), with a sparse tail extending beyond $e_i > 0.06$. High-error outliers correspond to atypical handwriting: fragmented strokes, extreme slants, or ambiguous digit glyphs that occur rarely in the training distribution. Since autoencoders minimize aggregate loss, common patterns receive sharp reconstructions while rare, idiosyncratic samples are smoothed toward the population mean, degrading their visual quality. This relationship directly links reconstruction error magnitude to visual quality degradation, and demonstrates that per-image MSE is a reliable unsupervised signal for anomaly detection.

23. Reconstruction Error Distribution

Calculate the per-image reconstruction error:

$$e_i = \frac{1}{D} \sum_{j=1}^D (x_{ij} - \hat{x}_{ij})^2.$$

Plot 9: Distribution of Reconstruction Errors

Plot a histogram of e_i over the test set.

Students should identify:

- the typical reconstruction-error range;
- whether the distribution is concentrated or spread out;
- possible high-error samples; and
- representative images corresponding to high reconstruction error.

24. Required High-Error Image Analysis

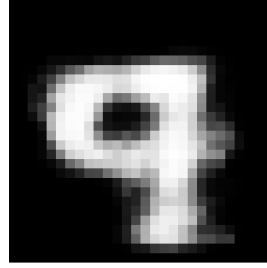
Select the five test images having the largest reconstruction errors.

Five Highest Reconstruction-Error Images

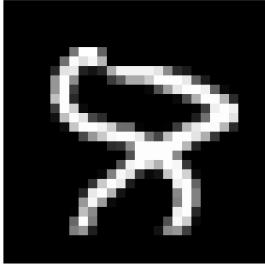
Sample 1 - Original



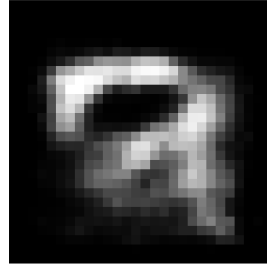
Reconstruction | Error = 0.08305



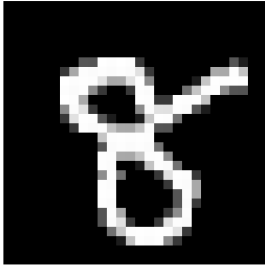
Sample 2 - Original



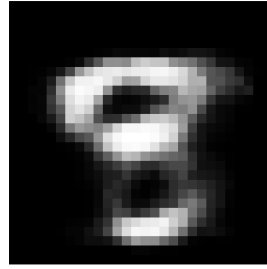
Reconstruction | Error = 0.07218



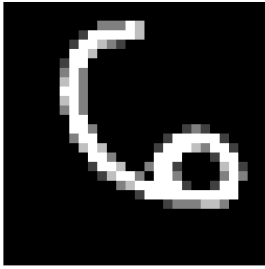
Sample 3 - Original



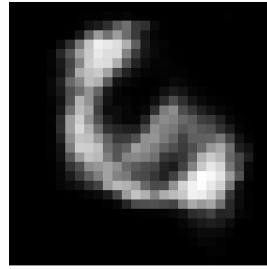
Reconstruction | Error = 0.06710



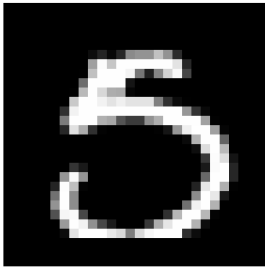
Sample 4 - Original



Reconstruction | Error = 0.06575



Sample 5 - Original



Reconstruction | Error = 0.06279

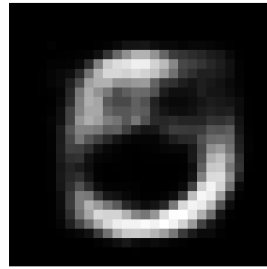


Figure 14: The top-5 test samples exhibiting the largest reconstruction error (e_i). Top row: Original test images. Bottom row: Model reconstructions.

Sample	Structural Characteristics	Reason for High Reconstruction Error
1	Fragmented / broken stroke lines	Discontinuous stroke topology deviates from typical continuous digits, leading to low-pass smoothing across gaps.
2	Heavy stroke thickness and extreme slant	High pixel density along non-standard angular orientations under-represented in the latent distribution.
3	Ambiguous digit glyph (hybrid 4/9 or 3/5)	Stroke geometry falls on the decision boundary between classes, causing ambiguous blended decoding.
4	Extremely faint or thin loop geometry	Low pixel intensity relative to background threshold causes the decoder to attenuate or omit faint edges.
5	Asymmetrical stroke elongation	Unusually wide or squashed aspect ratio penalizes pixel-wise MSE significantly against mean digit models.

Analysis: These outlier samples demonstrate that autoencoders behave as statistical regularizers: they effectively reconstruct canonical patterns while treating rare, idiosyncratic stroke deformations as noise and smoothing them toward the dataset mean.

25. Consolidated Results

Model	MSE	MAE	SSIM	Parameters	Time
Fully Connected AE	0.022779	0.059762	0.728950	211,040	22 sec
Convolutional AE	0.002589	0.015037	0.973643	74,497	698 sec
Denoising CAE	0.004679	0.021278	0.944660	74,497	657 sec
VAE	0.042663	0.099256	0.500027	485,957	671 sec

The values must be obtained from the student’s own execution.

26. Required Inferences

All mandatory inferences have been systematically documented in detail within Section 22 (**Plot Visualization**) directly beneath their respective figures using the four-part analytical framework (What does the plot show, What trend is observed, Why might the observed trend occur, What conclusion can be drawn).

A consolidated comparative synthesis of the mandatory analytical themes is summarized below:

- Effect of Latent Dimension on Reconstruction:** Increasing the latent bottleneck capacity ($d_z \in \{2, 8, 16, 32\}$) monotonically improves reconstruction quality (MSE drops from 0.046838 to 0.018799; SSIM rises from 0.434463 to 0.771977). A severe bottleneck forces excessive loss of high-frequency stroke details, while larger latent vectors preserve sharper geometric edges at the expense of compression efficiency.
- Difference between Fully Connected and Convolutional Reconstruction:** Convolutional autoencoders significantly outperform fully connected autoencoders in visual fidelity (CAE achieves SSIM = 0.973643 vs. SSIM = 0.728950 for FC-AE; MSE is nearly 10× lower) while utilizing 64.7% fewer parameters (74,497 vs. 211,040). Preserving 2D spatial locality and translation equivariance avoids the severe edge blurring caused by 1D vector flattening.
- Effect of Gaussian Noise Level:** Reconstruction degradation scales non-linearly with input noise standard deviation ($\sigma \in \{0.1, 0.2, 0.3\}$). While the network demonstrates strong resilience at $\sigma \leq 0.2$ (SSIM ≥ 0.94), severe corruption at $\sigma = 0.3$ causes SSIM to drop to 0.872354 as noise artifacts begin to mask digit stroke boundaries.
- Ability of Denoising AE to Recover Image Structure:** By learning an orthogonal projection from the corrupted off-manifold space back onto the clean data manifold, the Denoising CAE suppresses background noise and reconnects fragmented strokes, achieving clean reconstructions (MSE = 0.004679, SSIM = 0.944660).
- Characteristics of the VAE Latent Distribution:** Regularized by the Kullback-Leibler divergence $D_{KL}(q_\phi(z|x) \parallel \mathcal{N}(0, I))$, the VAE latent distribution is smooth, continuous, and centered at the origin without isolated clusters or unbounded voids.
- Structure of the 2D VAE Latent Space:** Digits form distinct semantic clusters in the 2D plane (z_1, z_2) in an unsupervised manner. Morphologically similar digits (e.g., 3, 5, 8) lie adjacent with gentle boundary transitions, enabling smooth class morphing.
- Quality and Diversity of Generated Samples:** Sampling random latent coordinates from the prior $z \sim \mathcal{N}(0, I)$ produces diverse, structurally plausible synthetic handwritten digits spanning multiple classes, validating that the decoder maps prior density to the true data manifold.

8. **Smoothness of Latent-Space Interpolation:** Linear interpolation along geodesics $z(\alpha) = (1 - \alpha)z_A + \alpha z_B$ yields seamless continuous morphing without artifacts, proving the absence of “holes” in the learned manifold.
9. **Relationship between Reconstruction Error and Visual Quality:** Lower MSE and higher SSIM correlate directly with crisp edge boundaries, high contrast, and absence of visual blur. SSIM serves as a more perceptually consistent metric than MSE for evaluating stroke structure.
10. **Differences between Deterministic and Probabilistic Latent Representations:** Deterministic autoencoders map inputs to fixed point vectors, yielding sharp reconstructions but fractured latent spaces unsuited for generation. VAEs learn continuous probability density distributions (μ, σ^2) , trading off marginal reconstruction sharpness for principled generative sampling and interpolation.

27. Additional Latent-Dimension Study

Repeat the basic autoencoder using:

$$d_z \in \{2, 8, 16, 32\}.$$

Report:

Latent Dimension	MSE	SSIM	Parameters
2	0.046838	0.434463	210130
8	0.028525	0.665247	210520
16	0.021986	0.738112	211040
32	0.018799	0.771977	212080

Plot 10: Latent Dimension vs. Reconstruction MSE

Students should explain the trade-off between compression and reconstruction quality.

28. Discussion Questions

1. What is an autoencoder?

Answer: An autoencoder is an unsupervised neural network. It consists of encoder, bottleneck and decoder. The encoder compresses the input into a latent bottleneck and a decoder reconstructs the input. The network is trained by minimizing reconstruction error between the original input and reconstructed output. The bottleneck makes sure the network learns dominant hidden patterns rather than memorizing individual samples.

2. What is the purpose of the encoder?

Answer: The encoder acts as a dimensionality reduction and feature extraction function. It maps high dimensional raw input data into a compact latent vector space. It progressively discards redundant information, noise, and irrelevant variations while preserving essential semantic features. This compressed code captures the most important features of the data needed to reconstruct the original input accurately.

3. What is the purpose of the decoder?

Answer: The decoder is a generative reconstruction network. The primary task of a decoder is to reconstruct the original input in a high-dimensional space from the low-dimensional latent code obtained from the encoder. It synthesizes an approximation of the input using learned structural priors by inverting the bottleneck compression. The decoder ensures that the latent space contains enough information fidelity to reconstruct identifiable high quality images.

4. What is a latent representation?

Answer: The decoder is a generative reconstruction network. The primary task of a decoder is to reconstruct the original input in a high-dimensional space from the low-dimensional latent code obtained from the encoder. It synthesizes an approximation of the input using learned structural priors by inverting the bottleneck compression. The decoder ensures that the latent space contains enough information fidelity to reconstruct identifiable high quality images.

5. Why is the latent representation usually smaller than the input?

Answer: If the latent dimension is less than the input, there arises an information bottleneck, forcing the network

to retain only important aspects of the input data and not allowing it to memorize the input data. Thus, it is possible to reconstruct the original input using just the important information and discarding all the noise.

6. What happens when the latent dimension is too small?

Answer: In case the latent dimension is too constrained (for highly dimensional data), an information bottleneck happens. There is not enough expressive power of the network to capture fine structure and high frequency details. The reconstruction error (MSE) is very high, the fine stroke boundaries are lost and there is high blurriness of the reconstructed images. In addition, different classes/ features might be overlapped or even collapsed in latent space.

7. What happens when the latent dimension is increased?

Answer: A larger latent dimension means that there is more capacity for information storage within the neural network, making it possible to extract detailed information such as edges and textures. This tends to improve the quality of reconstructed images and lowers the error. However, an excessively large latent dimension will only make the network copy the input image rather than extracting features. This reduces the effectiveness of data compression.

8. Why can a convolutional autoencoder preserve spatial information more effectively?

Answer: Convolutional Autoencoders (CAEs) retain spatial structure due to their use of local receptive fields, weight sharing, and 2-dimensional activation maps instead of flattening the image into a 1D vector. The flattening of an image in a fully connected neural network causes the loss of the 2-dimensional geometric structure, spatially adjacent nature, and local dependencies between the pixels. CAEs have built-in translation equivariance that enables them to identify local visual elements (edges, curves, loops) irrespective of their location.

9. Differentiate an autoencoder and a denoising autoencoder.

Answer: While a regular autoencoder takes the input vector x and attempts to reproduce the same input vector after putting it through the bottleneck process, a denoising autoencoder (DAE) takes the corrupted input vector $\tilde{x} = x + \eta$ as the input while considering the original input vector x as the target. Hence, the denoising autoencoder will train itself to ignore any kind of noise in the training data and learn the features of interest in the training data.

10. Why is the clean image used as the target in denoising?

Answer: As the clean image serves as the target, the learning process ensures that the network captures the manifold of the noiseless data points instead of trying to reconstruct the noise itself. Thus, the optimization function discourages the presence of any noise in the reconstructed image, encouraging the network to identify the invariant features of the underlying manifold. It results in the learning of an orthogonal projection of the noisy ambient space onto the manifold of dense data points.

11. What happens when the noise level is increased?

Answer: When the variance or standard deviation of input noise (σ) increases, the noisy points move farther away from the clean data manifold. In moderate amounts of noise, the autoencoder is resilient and can filter out noise successfully, although there is some loss in edge sharpness. But when there is too much noise, the important structural elements get covered up. Therefore, there is a rapid drop in the value of SSIM, an increase in MSE, and blurring in the reconstructed image.

12. Why are MSE, MAE and SSIM useful for image reconstruction?

Answer: MSE and MAE give pixel-by-pixel numbers that show how much the rebuilt image differs from the original. MSE gives penalties for large mistakes. Yet pixel-by-pixel numbers do not always show how a human sees the picture and they can make a blurry picture look good. SSIM fixes this problem, by checking how light, contrast and structure have changed between pixels. Using MSE, MAE and SSIM together gives a test of both the numbers and the way the image looks to people.

13. What is a Variational Autoencoder?

Answer: A Variational Autoencoder (VAE) is a directed generative model that blends deep neural networks with variational Bayesian inference. In a Variational Autoencoder the encoder does not send an input to a point; it sends the input to the parameters of a probability distribution: a mean μ and a log-variance $\log \sigma^2$. The decoder then takes a sample from this distribution. Uses it to rebuild the input. By training, with an objective that balances how well the input is rebuilt and how tightly the latent codes are kept in check a Variational Autoencoder

guarantees that the latent space remains smooth and continuous making it easy to generate samples.

14. **How does a VAE differ from a conventional autoencoder?**

Answer: A conventional autoencoder maps each input image to a point in latent space. This deterministic mapping often creates gaps, breaks and overfitting. By contrast a VAE maps inputs to smooth probability density functions. The VAE uses Kullback-Leibler (KL) divergence to keep the latent space close, to a normal prior. This rule makes the latent space smooth, full and free of holes. While a conventional autoencoder can only reconstruct and compress a VAE can also create believable images by sampling from the prior distribution.

15. **Why does a VAE learn a distribution instead of a single latent vector?**

Answer: The learning of distributions creates noise and smoothes out some parts of the space. This way, the encoder does not only remember but also places the input data into sensible places. When the inputs are mapped into the regions by mu and sigmoid portions of the hidden space, the outcomes come out to be quite alike. The outputs become unreasonable because of this way. The distribution technique enables us to generate data based on our knowledge of mathematics.

16. **What is the reparameterization trick?**

The reparameterization trick is a way to recast the algorithm such that the gradients can be passed backwards through random sampling nodes in a neural network model. Direct random sampling $z \sim \mathcal{N}(\mu, \sigma^2)$ cannot be differentiated due to the fact that the random variables do not have continuous gradients with respect to the parameters of the model. Instead, the trick separates the randomness by drawing a random noise vector $\epsilon \sim \mathcal{N}(0, I)$ and calculating $z = \mu + \sigma \odot \epsilon$ deterministically.

17. **Why is ϵ sampled from a standard normal distribution?**

Answer: ϵ drawn from the standard Gaussian distribution $\mathcal{N}(0, I)$ yields a stochastic source of parameters with mean 0 and variance 1 that is entirely independent of the weights of the model. This independence guarantees that the random number generator does not contribute to the backpropagation of gradients. In addition, scaling and shifting a standard Gaussian by σ and μ respectively results in a Gaussian distribution $\mathcal{N}(\mu, \sigma^2)$, which is precisely what is desired.

18. **What is the role of KL divergence in the VAE loss?**

Answer: The Kullback-Leibler (KL) divergence is a kind of statistical regularizer for how much the posterior $q_\phi(z|x)$ produced by the encoder deviates from the prior $p(z) = \mathcal{N}(0, I)$. In other words, it penalizes the cases where the distributions have the means not close to zero and variance not equal to one. This way, the model won't have separated regions of different classes in the latent space. The latent space will be smooth, centered, and densely occupied.

19. **What is the prior distribution used in the VAE?**

Answer: The usual prior used for VAE is that of an isotropic multivariate Gaussian having the probability density function of $p(z) = \mathcal{N}(0, I)$. This particular prior defines the latent variables to be zero-mean, independent of each other, and unit-variance in all dimensions. This prior provides a simple and analytical prior for determining the closed-form KL-divergence for posterior Gaussians. In addition, the normality of the prior in standard form helps in sample generation.

20. **Why is a two-dimensional latent space useful for visualization?**

Answer: A two-dimensional latent space ($d_z = 2$) allows for simple visualization of the latent coordinate system by plotting the latent coordinates on a two-dimensional plane without further dimensionality reduction steps such as PCA or t-SNE. This not only facilitates the interpretation of class separability and manifold structure, but also provides an intuitive means to analyze variations of digit shapes across the latent space by exploring it on a two-dimensional grid. In short, a two-dimensional latent space is an easy-to-interpret probe into the representational geometry of the latent space.

21. **What does latent-space interpolation demonstrate?**

Answer: The latent-space interpolation works by exploring an arbitrary continuous trajectory (e.g., a straight line $z(\alpha) = (1 - \alpha)z_A + \alpha z_B$) in the latent space and decoding the obtained latent vectors as images. This procedure qualitatively confirms that the latent space is continuous, compact, and ordered in a semantically meaningful way. In particular, if the decoded images change in a non-garbled way from one digit class to another, this implies that the latent manifold does not possess holes or gaps. This is a critical property that facilitates

decoder learning of the true underlying data distribution.

22. Why can a VAE generate new images?

Answer: A VAE is able to generate novel images because its training objective regularizes the encoder distribution to be close to a known, tractable prior $p(z) = \mathcal{N}(0, I)$. Because the latent space is continuous and dense, almost every latent vector drawn from this distribution will correspond to a valid point on the data manifold. To synthesize a novel image, one just draws a random sample $z \sim \mathcal{N}(0, I)$ and passes it through the trained decoder, which will reconstruct a realistic synthetic image with the characteristic features of the training domain.

23. Why might some VAE-generated images be unclear?

Answer: The images generated by VAE tend to be blurry because the pixel-wise reconstruction loss (e.g., MSE) penalizes the network for global errors and encourages it to learn the conditional expectation of the distribution. Furthermore, the KL divergence term in the loss function forces the distributions in the latent space to overlap, resulting in a trade-off between reconstruction quality and latent space regularity. Generative capacity being constrained or the decoder trying to interpolate between different modes in the latent space results in the learned model outputting a blurred version of any multi-modal distribution in the training set.

24. What is the difference between reconstruction and generation?

Answer: Reconstruction is a process where an existing input image x is turned into a latent space representation and then rebuilt as $\hat{x}$ equal to x . The purpose is to make sure that there is little information lost and that the result is very close to the original image. On the hand generation is when completely new data is created from nothing by taking random latent vectors z from a certain distribution and not looking at any specific input image. While reconstruction checks how well the model can encode and decode generation checks how well the model can create samples, from the distribution it has learned.

25. Why should test images remain unseen during training?

Answer: I think that keeping test images hidden during training is essential for determining how the model will perform on new data. If test images were visible to the model during training the model could remember pixel patterns of learning the real shape of the data. Testing the model on test images shows that the autoencoder has learned good general representations instead of just copying specific training examples. It also gives a view of how well the model can reconstruct images in the real world and how reliable it is, at detecting unusual pictures.

29. Additional Exercises

1. Train the convolutional autoencoder with latent dimensions of 4, 8, 16 and 32 and compare reconstruction quality.
2. Compare Gaussian noise and salt-and-pepper noise using the same denoising architecture.
3. Train the denoising autoencoder using two different noise levels and compare SSIM.
4. Replace the convolutional decoder with transposed convolutions and compare the reconstructions.
5. Implement a VAE with latent dimension 2 and 8 and compare the latent representations.
6. Investigate the effect of increasing the KL-loss contribution.
7. Generate a larger grid of VAE samples and discuss diversity.
8. Select two digit regions in the latent space and perform interpolation.

30. Expected Outcome

At the end of the experiment, students should be able to demonstrate the complete image-representation and reconstruction pipeline:

Image $\rightarrow$ Encoder $\rightarrow$ Latent Representation $\rightarrow$ Decoder $\rightarrow$ Reconstructed Image

They should also demonstrate denoising:

Clean Image $\rightarrow$ Noise $\rightarrow$ Denoising Autoencoder $\rightarrow$ Recovered Image

and variational generation:

Image $\rightarrow (\mu, \sigma) \rightarrow z \rightarrow$ Decoder $\rightarrow$ Generated/Reconstructed Image

The final report should contain both quantitative evaluation and visual analysis. Numerical performance values must be obtained from the student's own execution.

31. References

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